

In this paper, every Ring is Commutative Ring with  $1 \neq 0$ .

Thm. Let  $A_1, \dots, A_k$  be Ideals in  $R$ . Define a map

$$\mathcal{Q}: R \rightarrow R/A_1 \times \dots \times R/A_k: r \mapsto (r+A_1, \dots, r+A_k)$$

is a Ring Homomorphism with  $\ker \mathcal{Q} = A_1 \cap A_2 \cap \dots \cap A_k$ .

If for each  $1 \leq i < j \leq k$ ,  $A_i + A_j = R$  (comaximal), then  $\mathcal{Q}$  is onto and

$$\ker \mathcal{Q} = A_1 \cap \dots \cap A_k = A_1 A_2 \dots A_k.$$

Therefore,  $R/(A_1 \dots A_k) = R/(A_1 \cap \dots \cap A_k) \cong R/A_1 \times \dots \times R/A_k$ .

Proof. First, prove this for  $k=2$ . The map

$$\mathcal{Q}: R \rightarrow R/A_1 \times R/A_2: r \mapsto (r+A_1, r+A_2)$$

is just the natural projection for  $R/A_1, R/A_2$  elementwise, so ring-homomorphism.

Moreover,

$$\begin{aligned} \mathcal{Q}(r) = 0 &\Leftrightarrow r+A_1 = A_1 \text{ and } r+A_2 = A_2 \\ &\Leftrightarrow r \in A_1 \text{ and } r \in A_2 \end{aligned}$$

so  $\ker \mathcal{Q} = A_1 \cap A_2$ . Now, suppose that  $A_1 + A_2 = R$ . Since  $1 \in R = A_1 + A_2$ ,  
 $x+y=1$  for some  $x \in A_1$  and  $y \in A_2$ .

Therefore,  $\mathcal{Q}(x) = \mathcal{Q}(1-y) = (x+A_1, 1-y+A_2) = (0+A_1, 1+A_2)$ .

And similarly  $\mathcal{Q}(y) = (1, 0)$ . Now, given  $(r_1+A_1, r_2+A_2) \in R/A_1 \times R/A_2$ ,

$$\begin{aligned} \mathcal{Q}(r_2x+r_1y) &= \mathcal{Q}(r_2)\mathcal{Q}(x) + \mathcal{Q}(r_1)\mathcal{Q}(y) \\ &= \mathcal{Q}(r_2)(\bar{0}, 1) + \mathcal{Q}(r_1)(1, \bar{0}) \\ &= (\bar{0}, \bar{r}_2) + (\bar{r}_1, \bar{0}) = (r_1+A_1, r_2+A_2). \end{aligned}$$

Thus  $\mathcal{Q}$  is onto. And  $A_1 A_2 \subseteq A_1 \cap A_2$ , because: for  $a_i \in A_1, b_i \in A_2$ ,

$$a_1 b_1 + \dots + a_n b_n \in A_1 \text{ and } A_2 \text{ since } A_1, A_2 \text{ are Ideal.}$$

Meanwhile, if  $A_1 + A_2 = R$ , for any  $c \in A_1 + A_2$ ,

$$c = c \cdot 1 = c(x+y) = cx + cy \in A_1 + A_2.$$

So proved.

Thm. Let  $R$  be a Ring with identity  $1 \neq 0$ .

Let  $e \in R$  be an idempotent in  $R$  such that  $er=re$  for all  $r \in R$ .

Then,  $Re$  and  $R(1-e)$  are (two-sided) ideal of  $R$  and that

$$R \cong Re \times R(1-e)$$

Moreover,  $e$  and  $1-e$  are identity for  $Re$  and  $R(1-e)$ , respectively.

Proof. Notice  $e$  commutes with respect to all elements in  $R$ ,  $Re = eR$ .

and given  $r_1, r_2 \in R$ ,  $r_1e - r_2e = (r_1 - r_2)e \in Re$ , so  $Re$  is subgroup

and given  $r \in R$  and  $e \in Re$ ,  $re \in Re$ , so  $Re$  is left-sided ideal.

Since  $eR = Re$ , similarly it is right-sided ideal.

Moreover,  $1-e$  also commutes in  $R$ , so  $R(1-e)$  is ideal, similarly  $Re$ .

Given  $r \in R$ ,  $r = re + r - re$ , so  $R = Re + R(1-e)$  maximal.

By the Chinese Remainder Theorem,  $Re \cap R(1-e) = Re(1-e)$ .

observe that: given  $r_1, r_2 \in R$ ,

$$r_1e \cdot (r_2 - r_2e) = r_1r_2e - r_1r_2e^2 = r_1r_2(e - e^2) = r_1r_2 \cdot 0 = 0.$$

Therefore,  $Re \cdot R(1-e) = \{0\}$ . Meanwhile,

$$\mathcal{Q}: R \rightarrow Re; r \mapsto re$$

is onto Ring-homomorphism, because: Surjective is trivial and

$$\mathcal{Q}(r_1 + r_2) = (r_1 + r_2)e = r_1e + r_2e = \mathcal{Q}(r_1) + \mathcal{Q}(r_2) \text{ and}$$

$$\mathcal{Q}(r_1r_2) = r_1r_2e = r_1r_2e^2 = r_1e r_2e = \mathcal{Q}(r_1)\mathcal{Q}(r_2)$$

And  $\mathcal{Q}(r) = re = 0 \Leftrightarrow r \in R(1-e)$ , because

$$re = 0 \Rightarrow r = (r-e) - (r-e) \cdot e \in R(1-e)$$

$r \in R(1-e) \Rightarrow r = r' - r'e$  for some  $r' \in R \Rightarrow re = r'e - r'e^2 = 0$ . So  $\ker \mathcal{Q} = R(1-e)$ .

So,  $Re \cong R/R(1-e)$  and similarly  $R(1-e) \cong R/Re$ .

Consequently, the Chinese Remainder theorem gives

$$R \cong R/\{0\} \cong (R/R(1-e)) \times (R/Re) \cong Re \times R(1-e)$$

Further given  $re \in Re$ ,  $ree = ere = re^2 = re$ , so  $e$  is identity in  $Re$ , similarly  $1-e$  is identity of  $R(1-e)$ .  $\square$

Thm. Let  $R$  be a Ring with identity  $1 \neq 0$ .

Let  $e \in R$  be an idempotent in  $R$  such that  $er = re$  for all  $r \in R$ .

Then,  $Re$  and  $R(1-e)$  are (two-sided) ideal of  $R$  and that

$$R \cong Re \times R(1-e)$$

Moreover,  $e$  and  $1-e$  are identity for  $Re$  and  $R(1-e)$ , respectively.

Proof. Notice  $e$  commutes with respect to all elements in  $R$ ,  $Re = eR$ .

and given  $r_1, r_2 \in R$ ,  $r_1e - r_2e = (r_1 - r_2)e \in Re$ , so  $Re$  is subgroup

and given  $r \in R$  and  $r'e \in Re$ ,  $rr'e \in Re$ , so  $Re$  is left-sided ideal.

Since  $eR = Re$ , similarly it is right-sided ideal.

And, since  $(1-e)^2 = 1-e+e-e^2 = 1-e$ , so  $1-e$  is also idempotent and

$1-e$  also commutes with all elements in  $R$ , so  $R(1-e)$  is also ideal.

And, given  $re \in Re$ ,  $re \cdot e = re = e \cdot re$ , so  $e$  is identity of  $Re$ ,

and given  $r(1-e) \in R(1-e)$ ,  $r(1-e) \cdot (1-e) = r(1-e+e-e^2) = r(1-e) = (1-e)r(1-e)$ ,

so  $1-e$  is an identity of  $R(1-e)$ .

Define a map  $\mathcal{Q}: R \rightarrow Re \times R(1-e): r \mapsto (re, r-re)$ . Then,

$$\begin{aligned}\mathcal{Q}(r_1 + r_2) &= (r_1 + r_2)e, (r_1 + r_2) - (r_1 + r_2)e \\ &= (r_1e, r_1 - r_1e) + (r_2e, r_2 - r_2e) = \mathcal{Q}(r_1) + \mathcal{Q}(r_2) \\ \mathcal{Q}(r_1 r_2) &= (r_1 r_2 e, r_1 r_2 - r_1 r_2 e) \\ &= (r_1 e r_2 e, (r_1 - r_1 e)(r_2 - r_2 e)) = \mathcal{Q}(r_1) \mathcal{Q}(r_2)\end{aligned}$$

Therefore,  $\mathcal{Q}$  is Ring-Homomorphism and onto, because:

given  $(r_1e, r_2(1-e)) \in Re \times R(1-e)$ , let  $r = r_1e + r_2(1-e)$ . Then,  
 $\mathcal{Q}(r) = (r_1e^2 + r_2(1-e) \cdot e, r_1e(1-e) + r_2(1-e)^2) = (r_1e, r_2(1-e))$ .

Moreover,  $\ker \mathcal{Q} = \{r \in R \mid \mathcal{Q}(r) = (0, 0)\}$ , and  $re = 0$  and  $r(1-e) = 0$  implies

$re + r(1-e) = r = 0$ , so  $\ker \mathcal{Q} = \{0\}$ . So the first isomorphism theorem

gives that  $R / \ker \mathcal{Q} \cong R \cong Re \times R(1-e)$ .  $\square$

Prop. If  $R$  and  $S$  are non-zero rings, then  $R \times S$  is never a field.

Proof. Define a map

$$\mathcal{Q}: R \times S \rightarrow R: (r, s) \mapsto r.$$

Then,  $\mathcal{Q}((r_1, s_1) \cdot (r_2, s_2)) = \mathcal{Q}(r_1 r_2, s_1 s_2) = r_1 r_2 = \mathcal{Q}(r_1, s_1) \mathcal{Q}(r_2, s_2)$ ,  
and  $\mathcal{Q}((r_1, s_1) + (r_2, s_2)) = \mathcal{Q}(r_1 + r_2, s_1 + s_2) = r_1 + r_2 = \mathcal{Q}(r_1, s_1) + \mathcal{Q}(r_2, s_2)$ .

So,  $\mathcal{Q}$  is Ring Homomorphism and trivially onto.

Moreover,  $\ker \mathcal{Q} = \{(r, s) \in R \times S \mid r = 0\} = \{0\} \times S$ .

So,  $R \times S$  has non-trivial proper ideal, so it is never a field.  $\square$